Math Workout

Variables and calc

A rectangle is 12 cm wide and 5 cm tall. Area: 12 . 5 = 60 cm².

Double of 3.5 is 7.

Thousands still work: 100001.

Fractions, roots, powers

$\frac{2}{3}$

$\frac{x+1}{x-1}$

x^2

$(x+1)^n$

$\sqrt[2]{\{x\}}$

$\sqrt[3]{\{x+1\}}$

$\sqrt[3]{\{27\}}$

$\sqrt{16}$

$|x-5|$

Nested: $\frac{\sqrt{4}}{2^3}$

Trigonometry and logarithms

$\sin(x)$

$\cos(x)$

$\tan(x)$

$\cot(x)$

$\sec(x)$

$\csc(x)$

$\arcsin(x)$

$\arccos(x)$

$\arctan(x)$

$\sinh(x)$

$\cosh(x)$

$\tanh(x)$

$\coth(x)$

$\log(10)$

$\ln(2.718)$

Sums, products, limits

$$\sum_{i=1}^{n}(i)$$

$$\prod_{i=1}^{n}(i)$$

$\lim_{x\to 0}\left(\frac{\sin(x)}{x}\right)$

Integrals

$\int x^2$

$\int x^2 \,\mathrm{d}x$

$\int_0^1 x^2$

$\int_0^1 x^2 \,\mathrm{d}x$

$\oint_C^F l$

$\int_0^1 x^2$

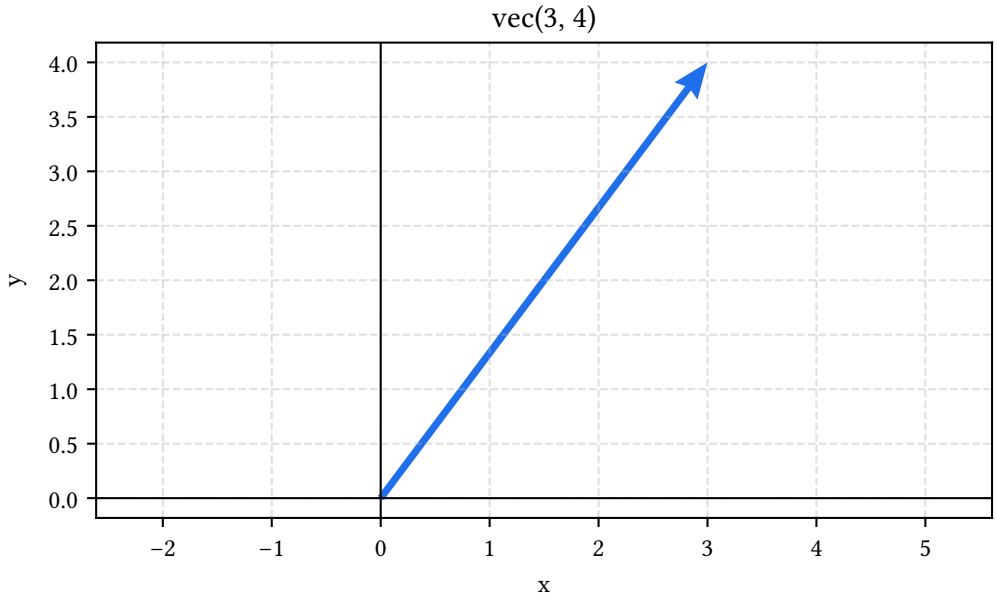
Piecewise functions and equation systems

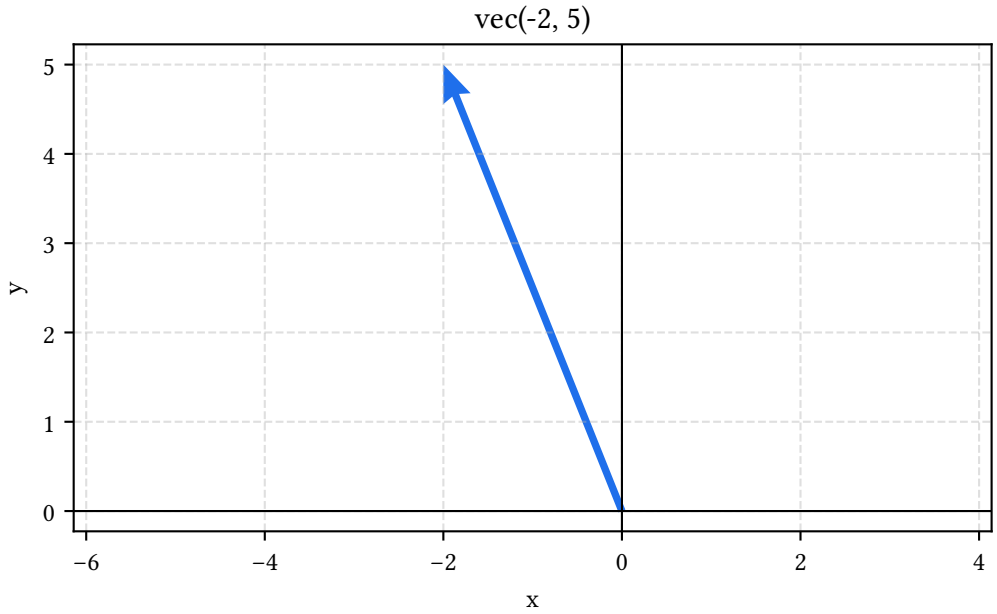
$$\begin{cases} x+y=3 \\ x-y=1 \end{cases}$$

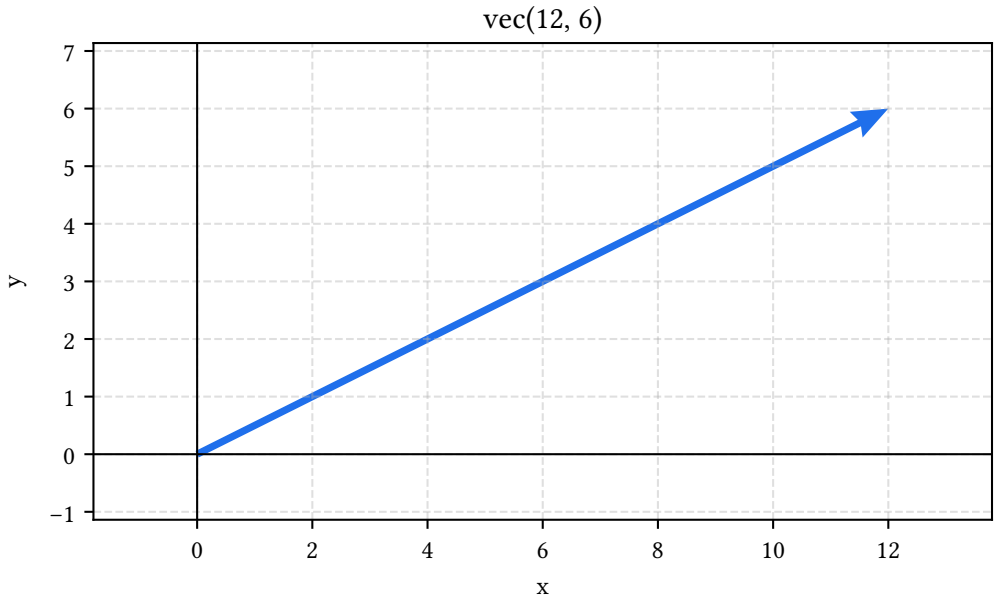
$$\begin{cases} xx>0 \\ 0x\leq 0 \end{cases}$$

$$\begin{cases} xx>0 \\ 0x\leq 0 \end{cases}$$

Vectors and binomials







$\binom{n}{k}$

$\binom{5}{2}$

Matrices and tables

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

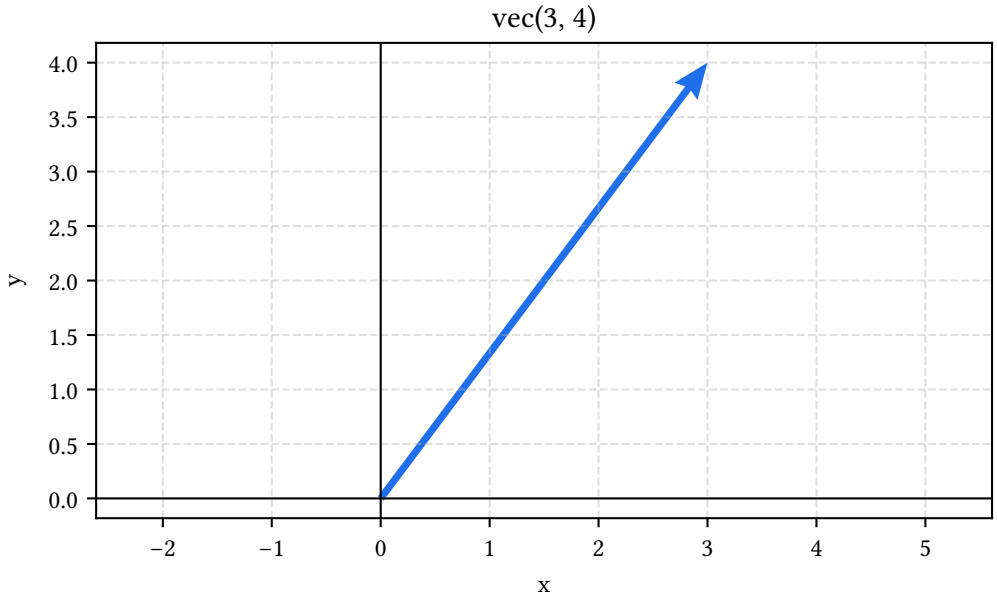
Name	Age
Alice	20
Bob	22

Inline math and subscripts

The solution is $x = 2$.

Discriminant: $b^2 - 4ac$

Corners A_1 and x_n, vector



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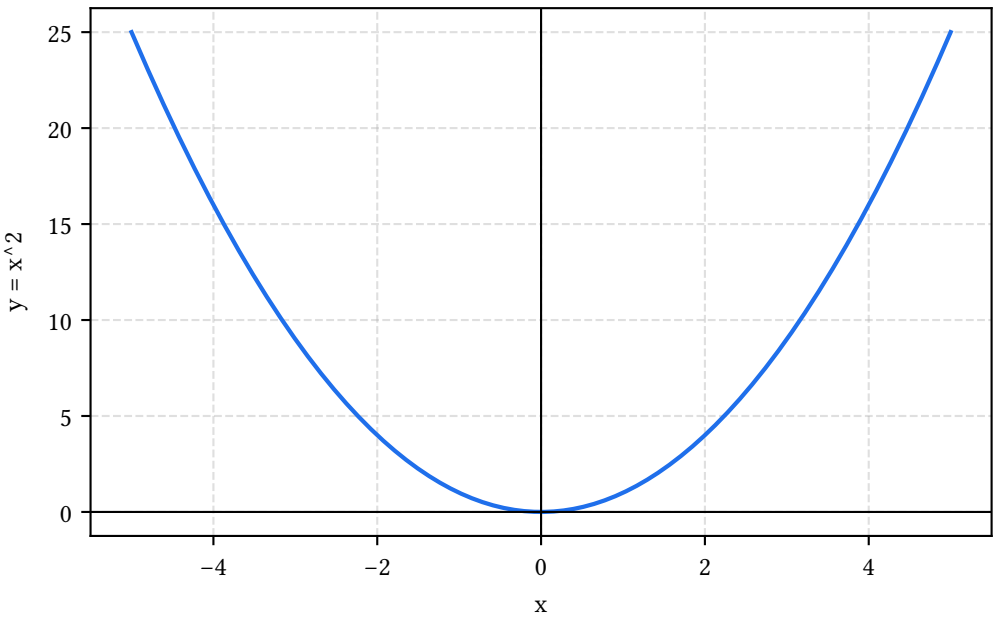
Spacing and symbols

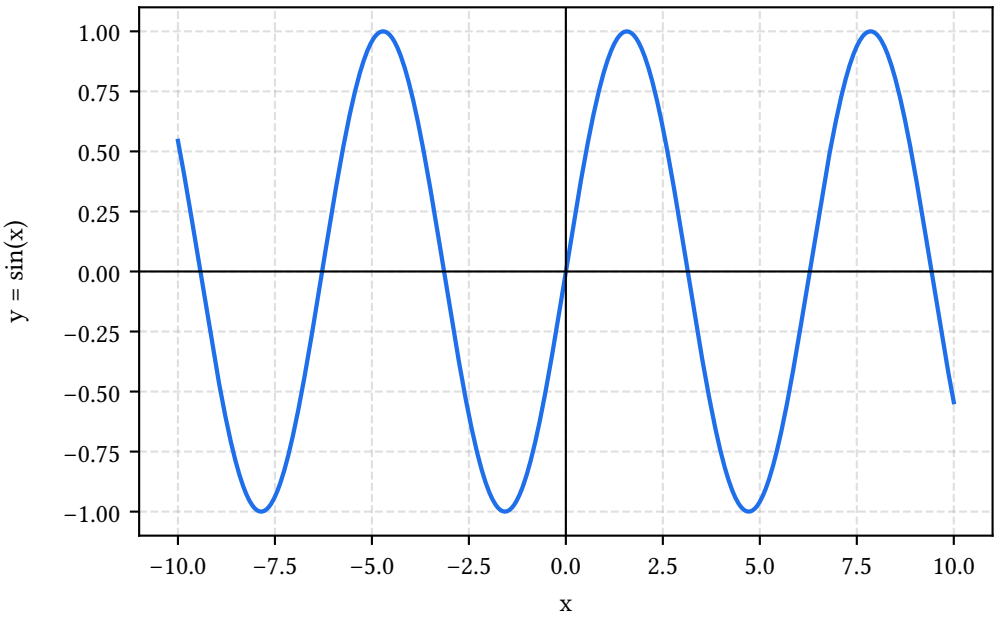
a quad b quad quad c thin d med e

α β γ $\pi\infty$

If x ≥ 0 and x ≤ 10 then x ≠ 5 ⟹ done.

Plots





Everything combined

$$\frac{\int_0^3 x^2}{\binom{4}{2}}$$